

AGILE DEVELOPMENT PROCESS AND DEVOPS LAB FAT

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Online Donation System

Problem Statement:

Ensure donations are processed and recorded accurately

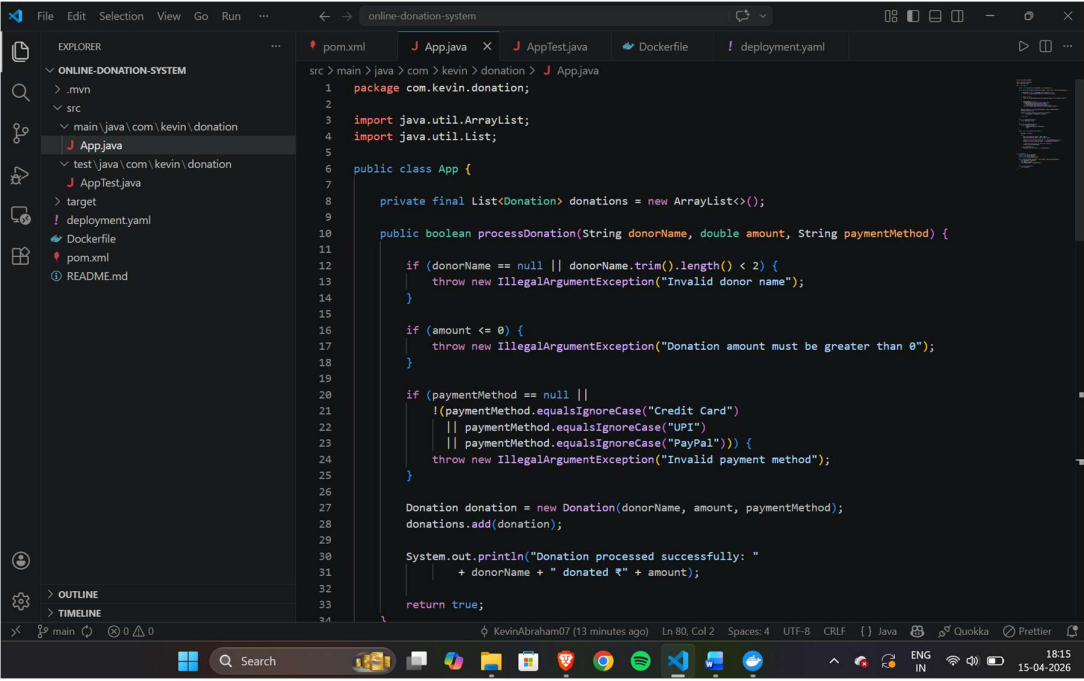
Tasks:

1. Develop donation processing logic
2. Write JUnit tests for transaction validation
3. Implement CI/CD pipeline with automated testing
 - Set up version control using GitHub
 - Configure CI pipeline to trigger on code commits
 - Automate build process using Maven
 - Integrate automated unit tests in pipeline
 - Configure CD pipeline for automatic deployment to staging/production
4. Deploy services on Docker and Kubernetes
 - Containerize application using Docker
 - Build and push Docker image to container registry
 - Create Kubernetes deployment and service YAML files

GitHub Repository Link: <https://github.com/KevinAbraham07/Agile-FAT.git>

App.java

Contains Java code for Online Donation System



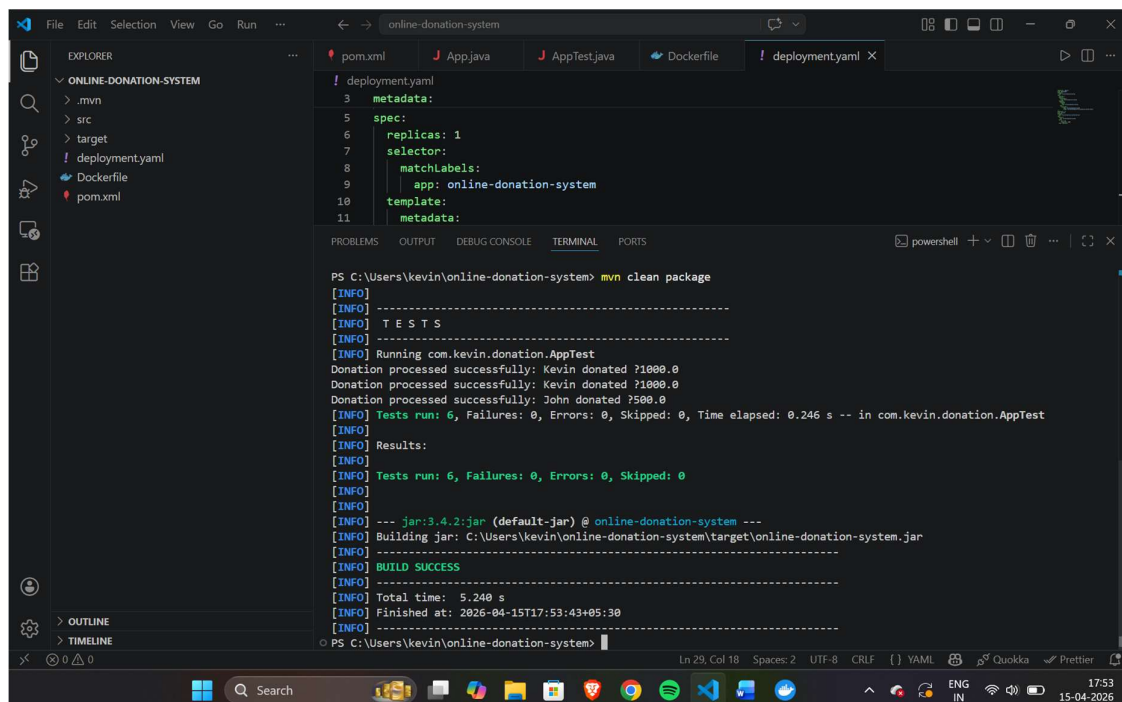
```
1 package com.kevin.donation;
2
3 import java.util.ArrayList;
4 import java.util.List;
5
6 public class App {
7
8     private final List<Donation> donations = new ArrayList<>();
9
10    public boolean processDonation(String donorName, double amount, String paymentMethod) {
11
12        if (donorName == null || donorName.trim().length() < 2) {
13            throw new IllegalArgumentException("Invalid donor name");
14        }
15
16        if (amount <= 0) {
17            throw new IllegalArgumentException("Donation amount must be greater than 0");
18        }
19
20        if (paymentMethod == null ||
21            !(paymentMethod.equalsIgnoreCase("Credit Card")
22              || paymentMethod.equalsIgnoreCase("UPI")
23              || paymentMethod.equalsIgnoreCase("PayPal"))) {
24            throw new IllegalArgumentException("Invalid payment method");
25        }
26
27        Donation donation = new Donation(donorName, amount, paymentMethod);
28        donations.add(donation);
29
30        System.out.println("Donation processed successfully: "
31            + donorName + " donated ₹" + amount);
32
33        return true;
34    }
35 }
```

Contains JUnit Testing Code for Online Donation System

Contains JUnit Testing Code for Online Donation System

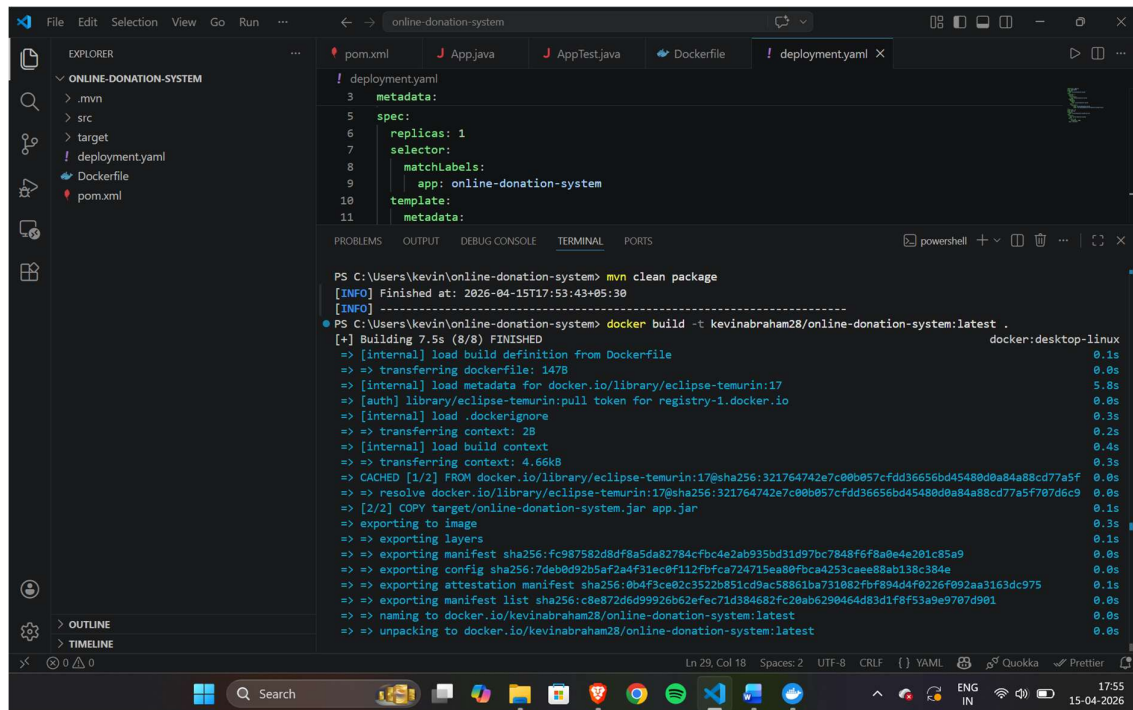


Compiles the Java application, runs tests, and packages it into an executable JAR file.



docker build -t kevinabraham28/online-donation-system:latest .

Builds a Docker image for the application and tags it for Docker Hub.



```
deployment.yaml
3  metadata:
4
5  spec:
6    replicas: 1
7    selector:
8      matchLabels:
9        app: online-donation-system
10   template:
11     metadata:
```

```
PS C:\Users\kevin\online-donation-system> mvn clean package
[INFO] Finished at: 2026-04-15T17:53:43+05:30
[INFO] -----
PS C:\Users\kevin\online-donation-system> docker build -t kevinabraham28/online-donation-system:latest .
[+] Building 7.5s (8/8) FINISHED
=> [internal] load build definition from Dockerfile                                docker:desktop-linux 0.1s
=> => transferring dockerfile: 147B                                              0.0s
=> [internal] load metadata for docker.io/library/eclipse-temurin:17             5.8s
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io            0.0s
=> [internal] load .dockerignore                                                  0.3s
=> => transferring context: 2B                                                  0.2s
=> [internal] load build context                                                  0.4s
=> => transferring context: 4.66kB                                              0.3s
=> CACHED [1/2] FROM docker.io/library/eclipse-temurin:17@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f 0.0s
=> => resolve docker.io/library/eclipse-temurin:17@sha256:321764742e7c00b057cfdd36656bd45480d0a84a88cd77a5f707d6c9 0.0s
=> [2/2] COPY target/online-donation-system.jar app.jar                         0.1s
=> => exporting to image                                                         0.3s
=> => exporting layers                                                         0.1s
=> => exporting manifest sha256:fc987582d8df8a5da82784cfbc4e2ab935bd31d97bc7848f6f8a0e4e201c85a9 0.0s
=> => exporting config sha256:7deb0d92b5af2a4f31ec8f112fbfca724715ea80fbca4253caee88ab138c384e 0.0s
=> => exporting attestation manifest sha256:0b4f3ce02c3522b851cd9ac58861ba731082fbf894d4f0226f092aa3163dc975 0.1s
=> => exporting manifest list sha256:c8e872d6d99926b62efec71d384682fc20ab6290464d83d1f8f53a9e9707d901 0.0s
=> => naming to docker.io/kevinabraham28/online-donation-system:latest          0.0s
=> => unpacking to docker.io/kevinabraham28/online-donation-system:latest       0.0s
```

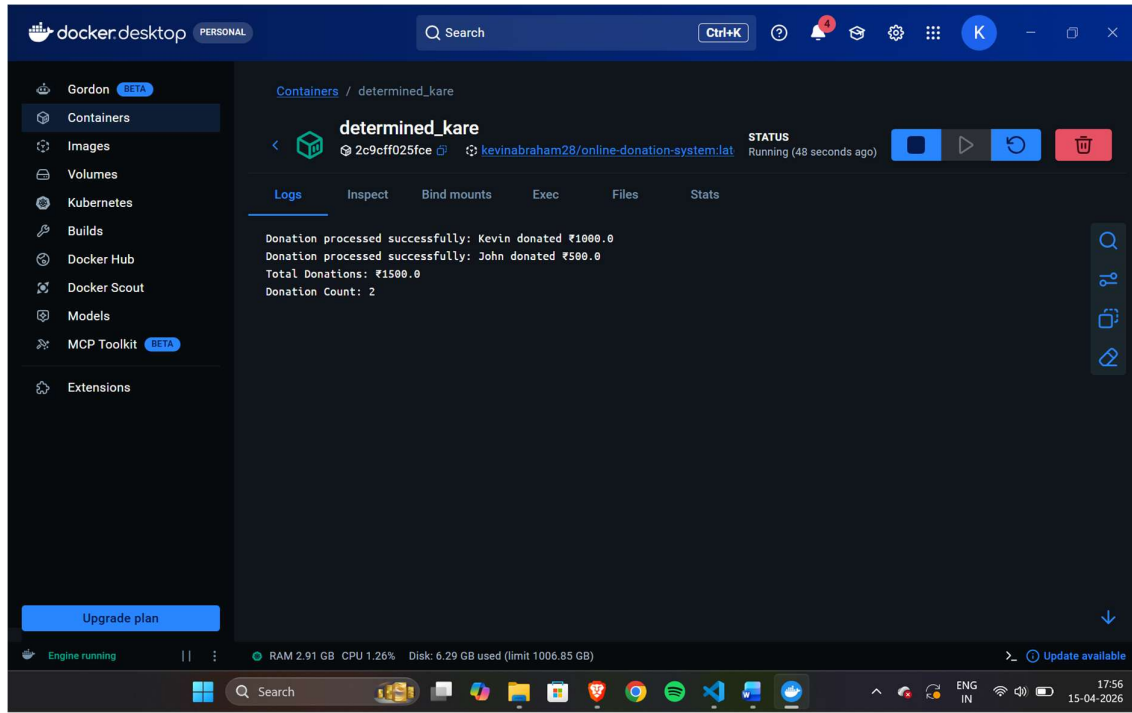
docker run kevinabraham28/online-donation-system:latest

Runs the built Docker image as a container locally for testing.

```
PS C:\Users\kevin\online-donation-system> docker run kevinabraham28/online-donation-system:latest
Donation processed successfully: Kevin donated ₹1000.0
Donation processed successfully: John donated ₹500.0
Total Donations: ₹1500.0
Donation Count: 2
```

New Docker Container Created

A Docker container was successfully created and run from the built image, therefore confirming the creation of cluster



docker push kevinabraham28/event-registration-app:latest

Uploads the Docker image to Docker Hub for remote storage and deployment.

```
PS C:\Users\kevin\online-donation-system> docker push kevinabraham28/online-donation-system:latest
The push refers to repository [docker.io/kevinabraham28/online-donation-system]
a1e6e1a167f1: Mounted from kevinabraham28/todo-management-app
d0061ec7f38e: Pushed
38d0b5bce0f2: Pushed
689b91d88a0f: Mounted from kevinabraham28/todo-management-app
a7428f4f5c1d: Mounted from kevinabraham28/todo-management-app
a149c623348d: Mounted from kevinabraham28/todo-management-app
f00bd8c488bb: Mounted from kevinabraham28/todo-management-app
latest: digest: sha256:c8e872d6d99926b62efec71d384682fc20ab6290464d83d1f8f53a9e9707d901 size: 856
```

kubectl apply -f deployment.yaml

Deploys the application in the Kubernetes cluster using the deployment configuration.

```
PS C:\Users\kevin\online-donation-system> kubectl apply -f deployment.yaml
deployment.apps/online-donation-system created
service/online-donation-system-service created
```

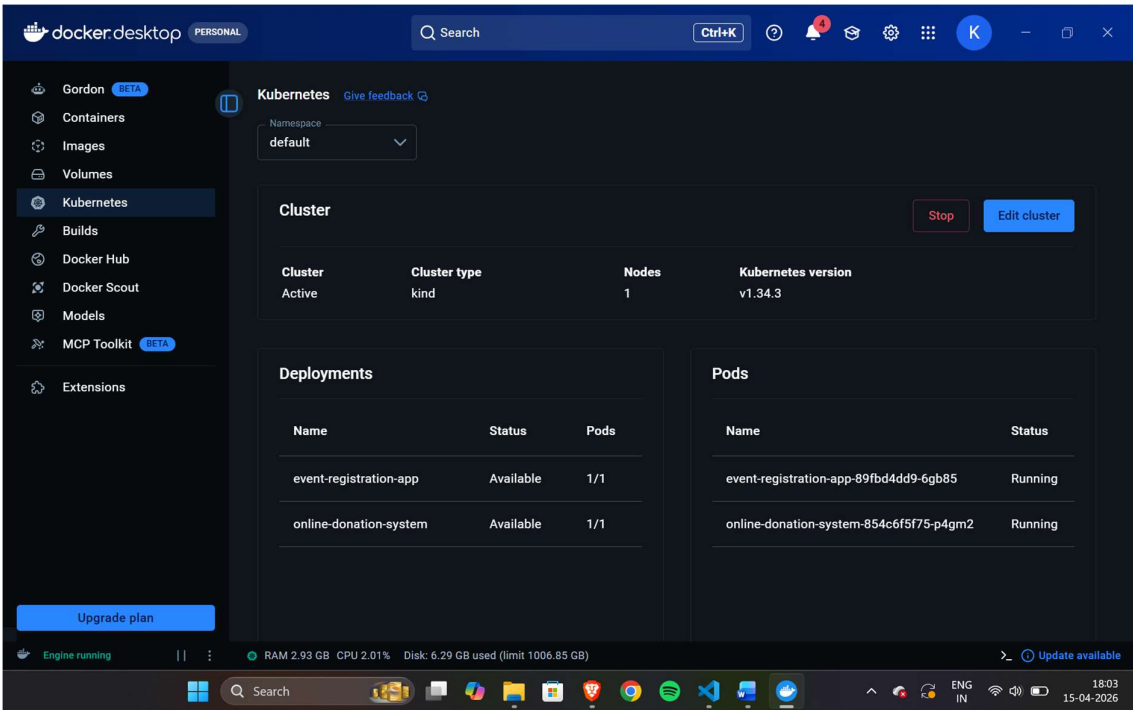
kubectl get pods

Displays the status of all Kubernetes pods in the cluster to verify that the deployed application container is running successfully.

```
PS C:\Users\kevin\online-donation-system> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
event-registration-app-89fbd4dd9-6gb85 1/1     Running   1 (58m ago) 4d19h
online-donation-system-854c6f5f75-p4gm2 1/1     Running   0           22m
```

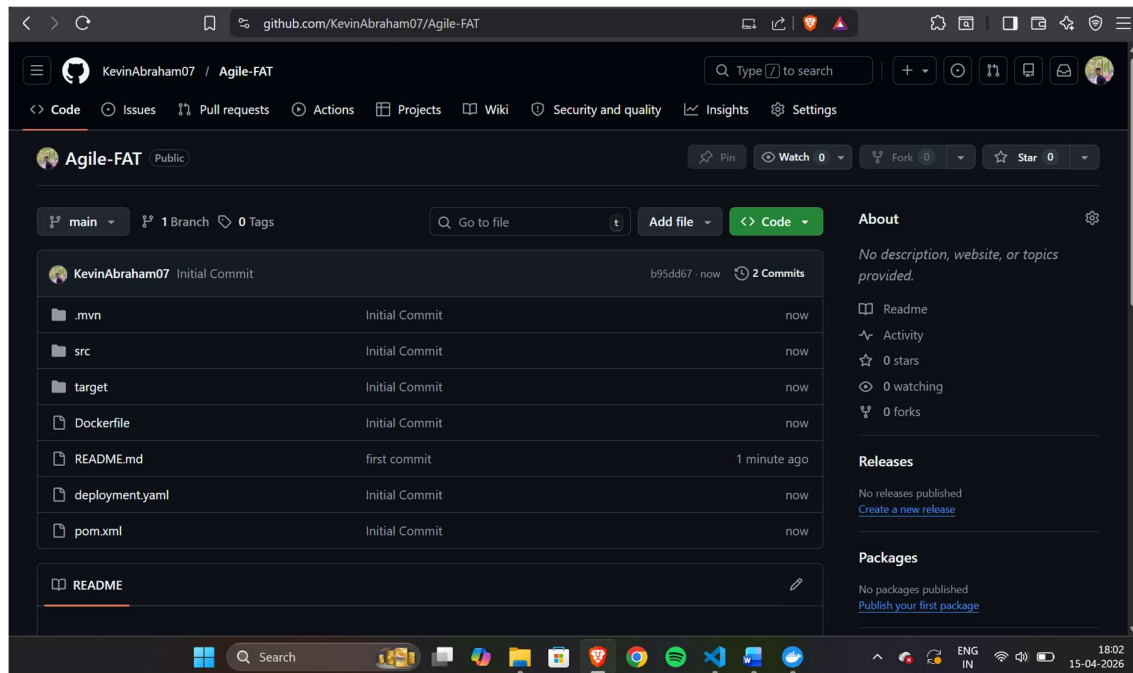
Pushed to Kubernetes Cluster

The application container is deployed to the Kubernetes cluster, where the deployment status is monitored.



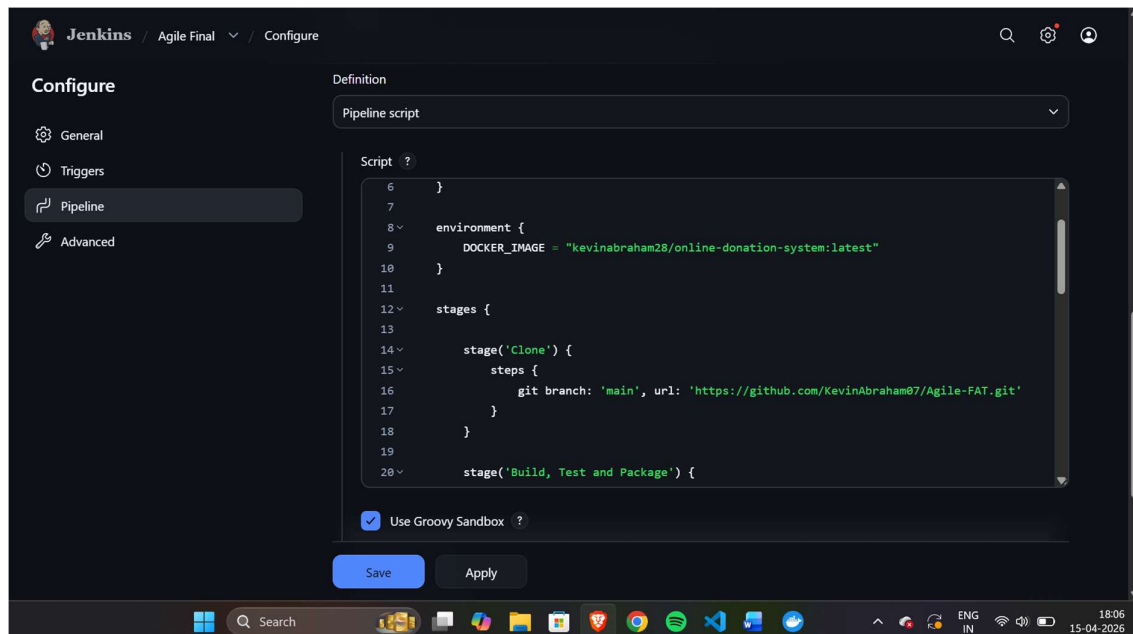
Push to GitHub Repository

The files are uploaded to GitHub to facilitate cloning in Jenkins



Jenkins Pipeline

The Jenkins pipeline configuration automates the CI/CD workflow by cloning the repository, building the project, running tests, containerizing the application, pushing the image, and deploying to Kubernetes.



CI/CD Pipeline Stages

The automated CI/CD pipeline is successfully executed

